NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Sorting Trays After New Labels

AP Statistics · Unit 3 · Topic 3.7 · B: 2026-12-07 · E: 2027-01-13

[STAR] TODAY'S PROBLEM

A cafeteria served 5,000 September trays after new sorting-bin labels were installed. The director claims at least 60% are sorted correctly. An SRS of 250 trays finds 136 correct; the club tests at $\alpha = 0.05$.

Imani: "The p-value is 0.0354, so reject H_0. The labels caused sorting below 60 percent for every month."

Drew: "The 5.6-point gap may be too small to matter, so it is not significant. Accept the director's claim."

The test conditions have been checked: random sample; $250 < 500$; null-expected counts 150 and 100.

September study

Source	Trays	Correct	α
September	5,000	claim: $\geq 60\%$	–
SRS	250	136	0.05

FIRST TAKE — before the video (5 min)

Can these September trays tell us what caused the result in every month? Give one reason from the study.

[BOOK] MAKE THE DECISION, THEN BOUND THE CLAIM (keep on the page)

For a one-proportion test, $z = (\widehat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$; for $H_a : p < p_0$, use the left normal tail. If the p-value is at most the predetermined α, reject H_0; otherwise fail to reject it. State evidence for H_a in context. Failing to reject does not establish that H_0 is true. Statistical significance uses the evidence rule; practical importance asks whether an effect is large enough to matter. A random sample supports inference to its population. A cause requires a comparison with random assignment.

Finish after the video:

DOK 1 **(a)** Define p and state the hypotheses for checking whether correct sorting is below 60%.

DOK 2 **(b)** Calculate $\widehat{p}$, the one-proportion z statistic, and the lower-tail p-value. Use the verified conditions above.

DOK 3 **(c)** In 4–5 sentences, correct both reports. Compare the p-value with α, separate statistical significance from practical importance, and limit the conclusion to the people or items studied and what the design can show about cause.

Frame: Because p-value = ________ is _______ α, I would __________ H_0 and conclude ______________________________.

Frame: The result is statistically __________, but the design does not show ______________________________; the 5.6-point gap's practical meaning depends on ____________________.

EXIT — turn this in

One thing the video changed about my first take: ____